

COVERAGE WHILE ON LEAVE • STUDY CP-101

A month of trial monitoring — automated, in SAS and R

The lead biostatistician is in Europe for a month. Monitoring keeps running on validated, deterministic SAS/R inside the existing environment — the covering colleague gets the right updates, escalation is automatic, and the manager gets peace of mind. No AI, no new tools, no PHI egress.

The third companion pipeline — the one you can ship today. • Illustrative mockups.

Scheduler

02:00 • 06:00 • 07:00 • Fri



SAS / R program

deterministic checks



Report

ODS / R Markdown



Notify

digest + urgent alert

Schedule the already-validated programs, hand over the runbook

⋮ Scheduled jobs — service account SVC-BIOSTAT (not a personal login)

02:00 daily

Data refresh + integrity & FRESHNESS gate

ingest latest SDTM/safety/IXRS; verify feeds updated

06:00 daily

Safety scan

AE/SAE • Hy's-Law/eDISH • QTcF • labs • DLT tally

07:00 daily

Routine digest → support

GREEN/AMBER summary + heartbeat

Fri 16:00

Weekly status + exec one-pager

→ support (full) • → manager (1 page)

on-event

Threshold ALERT

KRI/QLT/DLT/SAE → urgent email + escalation

⋮ Coverage runbook (PDF) — to support, cc manager

CP-101 — Monitoring coverage

 **GREEN** all checks within limits — no action

 **AMBER** an operational/quality KRI crossed — support reviews & acts

 **RED** safety threshold (DLT/SAE/Hy's-Law/QTcF) — ack ≤30 min, CONTACT MEDICAL MONITOR

Recipients primary + BACKUP colleague + role alias (never one person)

Escalation Medical Monitor: Dr. R. Okafor • +1-...

Honest note *a flag = a number crossed a line; it is NOT an interpretation. Humans decide.*

0

Nothing here is new code — the lead's normal monitoring programs are simply moved from “run when I remember” to “run on a clock,” under a service account that outlasts the lead's login.

The daily digest — GREEN — and the heartbeat that proves it ran

 Outlook — Daily Safety Digest

Daily Safety Digest — CP-101 — Tue 09 Jun

● GREEN

to: Support (covering biostatistician) • auto-sent 07:00

● Enrolled / dosed	24 / 24 across cohorts 1-4 • cohort 5 screening
● New AEs / SAEs since last run	3 AE (all Grade 1) • 0 SAE
● Hy's-Law / eDISH • QTcF	0 flags • 0 flags
● Labs / vitals out of range	0 clinically-notable
● Cohort DLT tally	cohort 4: 0/6 — within 3+3
● Open queries / overdue visits	11 open (none >30 d) • 0 overdue

✓ HEARTBEAT ran 07:00:04 • data fresh to 06:14 today • 28,114 records • SYSCC=0 • next run 10 Jun 07:00

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“No news” is never assumed: every run emits a heartbeat with a data-freshness timestamp — so a missed job or a frozen feed announces itself instead of hiding behind a green screen.

An enrollment KRI dips — AMBER, in the routine channel

⋮ Outlook — Daily Safety Digest

Daily Safety Digest — CP-101 — Tue 16 Jun

● AMBER

to: Support · auto-sent 07:00

● Enrollment vs plan — KRI BREACHED

76% of planned-to-date (KRI floor 80%) · screen-failure cluster at Site 02 (5 of last 7) · cohort-5 fill at risk

● Safety (AE/SAE · Hy's-Law · QTcF · DLT)

all within limits

● Labs / vitals

0 notable

● Queries / deviations

14 open · deviation trend flat

Runbook → AMBER: Support reviews enrollment, raises a site-activation action with the PM. Stays in the digest channel. The manager's Friday one-pager will show: flagged → being handled.

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An operational signal correctly stays in the routine channel — it does not cry wolf at the medical monitor. The support acts; the manager simply sees it was caught and handled.

A candidate Hy's-Law signal trips the URGENT tier — outside the digest

Outlook — URGENT safety alert

RED ALERT — candidate Hy's-Law lab pattern — Subject 0312 (cohort 5, top dose)

auto-sent 06:07 • to: Support + BACKUP + role alias • cc: PM

Deterministic criteria met: ALT $4.1\times$ ULN AND total bilirubin $2.3\times$ ULN (ALP $1.2\times$ ULN) — hepatocellular pattern, same episode.

- ▶ **CONTACT THE MEDICAL MONITOR NOW** — Dr. R. Okafor, +1-...
- ▶ Acknowledge this alert within 30 minutes (reply to confirm receipt).
- ▶ Evidence packet for Subject 0312 attached (see next page).

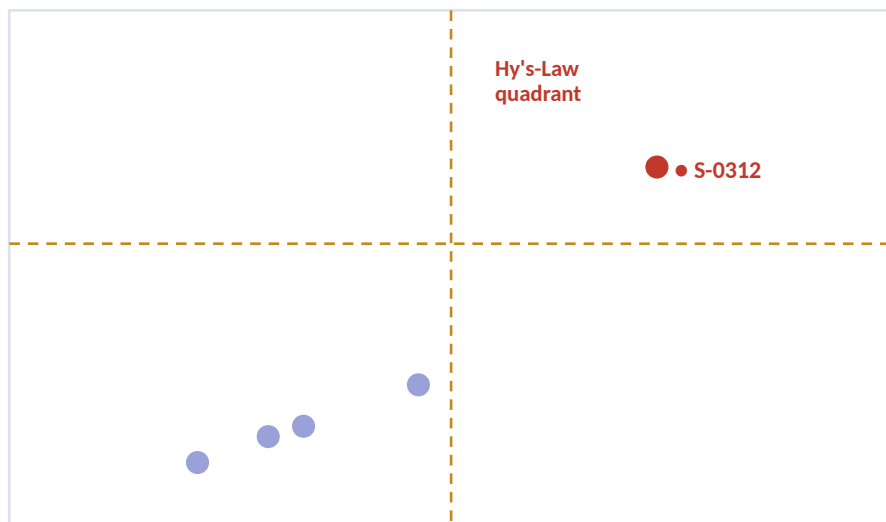
This is a SCREENING flag that matches Hy's-Law lab criteria — it is NOT a diagnosis. The medical monitor adjudicates.

CP101_S0312_eDISH_evidence.pdf · S0312_lab_trajectory.pdf · heartbeat: ran 06:07, data fresh to 05:51

The evidence is pre-packaged; the humans decide

Evidence packet — Subject 0312 (auto-generated, reproducible)

eDISH — peak ALT/ULN vs peak TBili/ULN



x: peak ALT (xULN) → y: peak total bilirubin (xULN) ↑

Packet also includes: lab trajectory (ALT/AST/TBL/ALP by visit) • dosing & exposure timeline • concomitant meds • the exact threshold rule that fired.

Escalation log (time-stamped, automatic)

06:07 ALERT fired — RED — S-0312

06:09 delivered: support, backup, alias, PM

06:31 support ACK (within 30-min SLA)

07:10 medical monitor reviewed packet

07:25 MM: repeat LFTs + hold cohort 5 dosing

Fri confirmatory labs trending down

Sun MM: not confirmed Hy's Law; cohort released

alert auto-stood-down; loop closed

System did 3 deterministic things: DETECTED • PAGED • PACKAGED. Every clinical decision was the medical monitor's.

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Under stress, the covering colleague isn't scrambling to pull data at 6 a.m. — it's already assembled, identically formatted, and reproducible. Detect, page, package; humans decide.

A complete, time-stamped, reproducible audit trail — nothing to reconstruct

Monitoring archive — CP-101 — the month, on the validated share

 logs/2026-06-*.log 22 dated SAS/R run logs — every job, every day, SYSCC=0

 digests/daily_*.pdf 22 daily safety digests (heartbeat on each)

 digests/weekly_*.pdf 4 weekly status packs + 4 manager one-pagers

 alerts/2026-06-16_AMBER_enrollment.pdf Week-2 enrollment KRI — flagged → handled (PM action)

 alerts/2026-06-25_RED_S0312_HysLaw.pdf Week-3 candidate Hy's-Law — escalated → adjudicated → stood down

 evidence/CP101_S0312_eDISH_evidence.pdf auto-built evidence packet (eDISH + trajectory + dosing)

 heartbeat/heartbeat_history.csv every run accounted for — no silent gaps in the month

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Because the whole system is deterministic and validated, the audit trail is automatic and trustworthy — the lead reconstructs nothing, and the handback is clean.

Four safeguards — because “no news” must never mean “good news”

Heartbeat / dead-man's-switch

Every successful run pings “ran OK, SYSCC=0”; an independent watcher on a second host escalates if the ping is missing — so a dead job announces itself.

Data-freshness gate (first check)

Before any safety logic runs, compare each feed's newest timestamp to now; if a source stopped refreshing, raise it — a clean green on stale data is worse than red.

Alert de-duplication

A persistent seen-flags table keys each finding by subject+test+event, so one real event alerts once — no storm, no muted thread burying the next real signal.

Primary + backup recipients

Never one person: a primary, a backup, and a role-based alias, with an acknowledgement SLA — so a 6 a.m. alert never lands in a void while everyone's away.

The SOP rule: “no news” is never interpreted as “all clear” — a successful, data-fresh run must say so explicitly, and the absence of any message is itself escalated.

Necessary, not sufficient — and that's exactly the point.

The lowest-risk pipeline — ship today

No AI, no new vendor, no PHI egress, no new validation surface. It reuses already-validated SAS/R inside the existing environment, so there's no new approval gate between today and go-live.

Deterministic & reproducible

Same data → byte-identical result, every run. Nothing to “explain,” no model drift — the audit trail is automatic and trustworthy.

Humans — and the medical monitor — still decide

It detects, pages, and packages pre-specified threshold crossings. It does not triage, interpret, or adjudicate. A flag is a prompt for a human, never a finding.

The seam the AI hybrid fills later

Narrative triage, cross-signal summarization, plain-language briefing — that's the next layer. Ship this now; add AI on top when it's approved.

Peace of mind: monitoring never stops, even silence is verified (heartbeat + data-freshness), and every escalation is automatic and logged — while the lead is unreachable in Europe.